

Discrete Mathematics – Spring 2026

Homework 10: Number Theory

Due Sunday April 19 @11:59:00pm Eastern

As always, show your work.

1. Divisibility

What are the quotient and remainder when:

- 1,001 is divided by 23
- -512 is divided by 15

Show your work.

2. Prime Factorization

Find the prime factorization of each of the following integers. Show all your steps.

- 360
- 2,310
- $11!$ (*Hint: Rather than computing $11!$ directly, think about how many times each prime $p \leq 11$ appears as a factor across all integers from 1 to 11.*)

3. GCD and LCM

Consider the two positive integers:

$$a = 2^3 \times 3^4 \times 5^2 \times 11^3 \quad \text{and} \quad b = 2^5 \times 3^2 \times 7^2 \times 11^1$$

Using the GCD formula (not Euclid's algorithm), find $\gcd(a, b)$ and $\text{lcm}(a, b)$. Verify that $\gcd(a, b) \times \text{lcm}(a, b) = a \times b$. Show all your steps.

4. Extended Euclid Algorithm

Does the multiplicative inverse of 103 in $\mathbb{Z}_{128}$ exist? If so, find it using the extended Euclid algorithm. If it does not exist, state why. Show all your steps.

5. Euclid's Algorithm and Bézout's Theorem

- Use Euclid's algorithm to compute $\gcd(540, 168)$. Show all steps.
- By Bézout's theorem, there exist integers x and y such that $540x + 168y = \gcd(540, 168)$. Using back-substitution from part (a), find such integers x and y .

6. Co-primality

Let $a, b \in \mathbb{Z}$. We say a and b are **relatively prime** (or **coprime**) iff $\gcd(a, b) = 1$.

Show that for every integer $k \geq 1$, the numbers $3k + 2$ and $4k + 3$ are relatively prime.

